AWS Systems Manager Based Disk Utilization Discovery for EC2 Instances

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1. Introduction

Organizations running workloads on Amazon EC2 often manage multiple instances across different environments. Monitoring disk utilization is essential to ensure application availability, prevent storage-related issues, and support capacity planning. Traditionally, administrators would connect to each server using SSH or RDP to check storage utilization, which becomes difficult and time-consuming at scale.

AWS Systems Manager provides a secure and centralized method to manage EC2 instances without requiring direct server access. Using Systems Manager Run Command, administrators can remotely execute commands and retrieve operational information from EC2 instances.

This document demonstrates how disk utilization can be retrieved from an EC2 instance using AWS Systems Manager, explains the AWS components involved, highlights benefits and limitations, and provides implementation screenshots and customer communication examples.

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2. Objective

The objective of this exercise is to:

- Retrieve disk utilization information from an EC2 instance using AWS Systems Manager.
- Avoid direct SSH/RDP access to servers.
- Understand the AWS resources involved in the process.
- Demonstrate secure remote command execution.
- Evaluate benefits and limitations of the solution.
- Explore alternate approaches for collecting disk utilization information.

3. Solution Architecture

[Insert Architecture Diagram]

Administrator
|
v
AWS Systems Manager
|
v
Run Command
|
v
SSM Agent
|
v
EC2 Instance
|
v
df -h Command
|
v
Disk Utilization Output

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4. AWS Components Used

4.1 Amazon EC2

Amazon EC2 provides the virtual server on which the workload is running. Disk utilization is collected from this instance.

4.2 AWS Systems Manager

AWS Systems Manager is a management service that enables administrators to remotely execute commands and manage EC2 instances without requiring SSH or RDP access.

4.3 SSM Agent

The SSM Agent runs inside the EC2 instance and communicates with AWS Systems Manager. It receives commands, executes them locally, and returns the results.

4.4 IAM Role

An IAM role was attached to the EC2 instance to grant permissions required for Systems Manager communication.

Role Used:
EC2-SSM-Role

Policy Attached:
AmazonSSMManagedInstanceCore

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5. Implementation Steps

Step 1: Launch EC2 Instance

Created an Amazon Linux EC2 instance for testing.

Screenshot 1

EC2 Instance Running

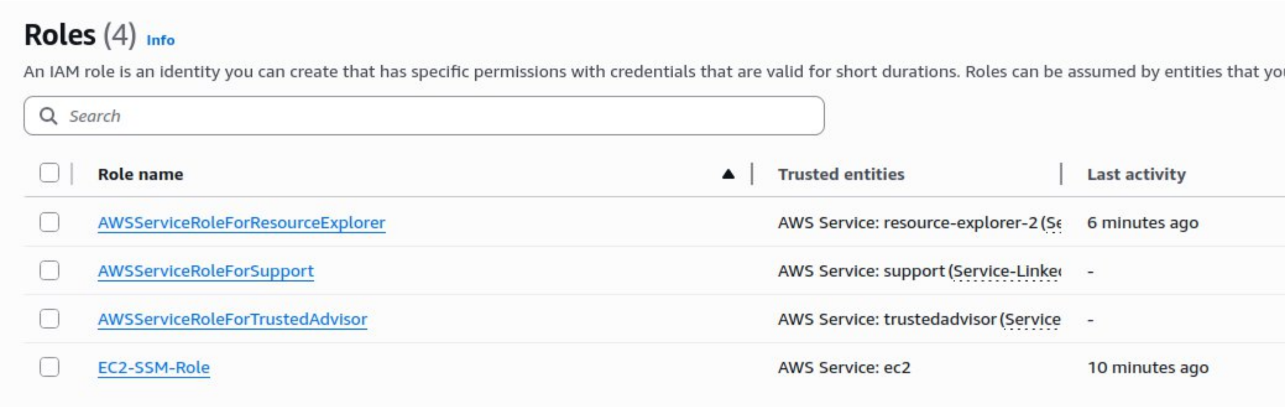
Step 2: Create IAM Role

Created IAM role named EC2-SSM-Role.

Attached policy:
AmazonSSMManagedInstanceCore

Screenshot 2

IAM Role Created [EC2-SSM-Role]

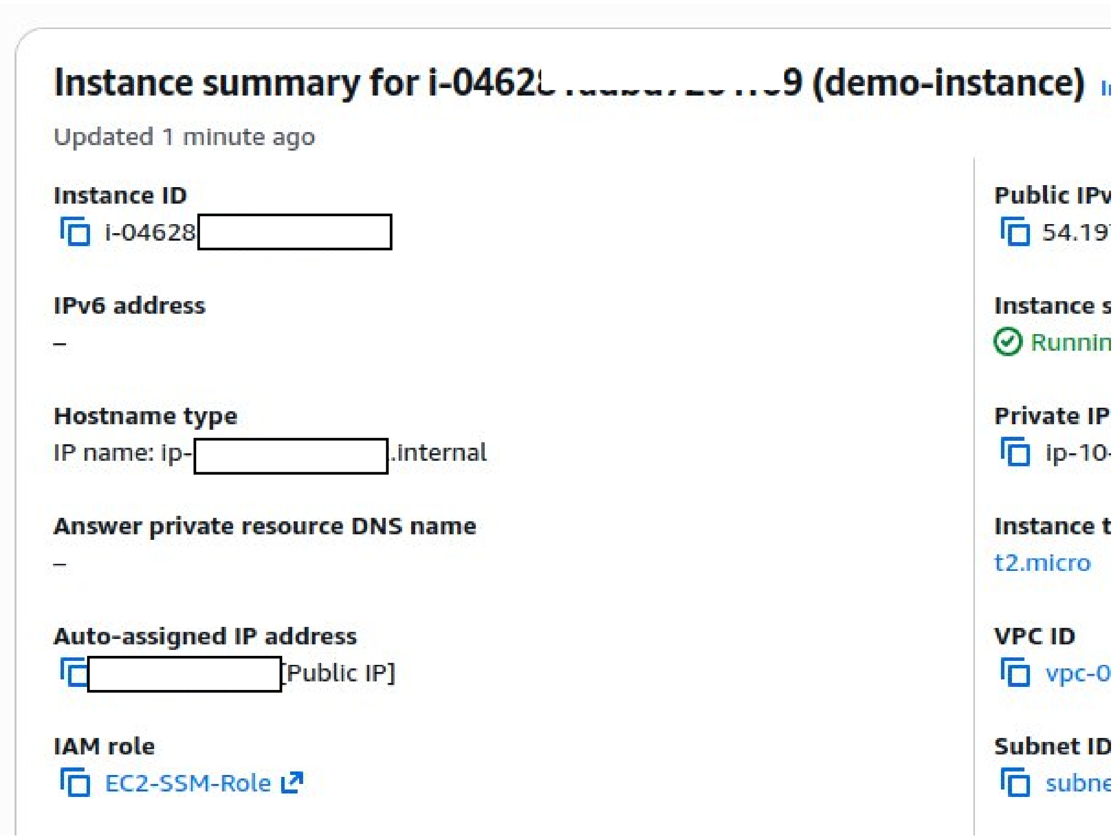


Step 3: Attach IAM Role to EC2 Instance

Attached the IAM role to the EC2 instance.

Screenshot 3:

IAM Role Attached to EC2

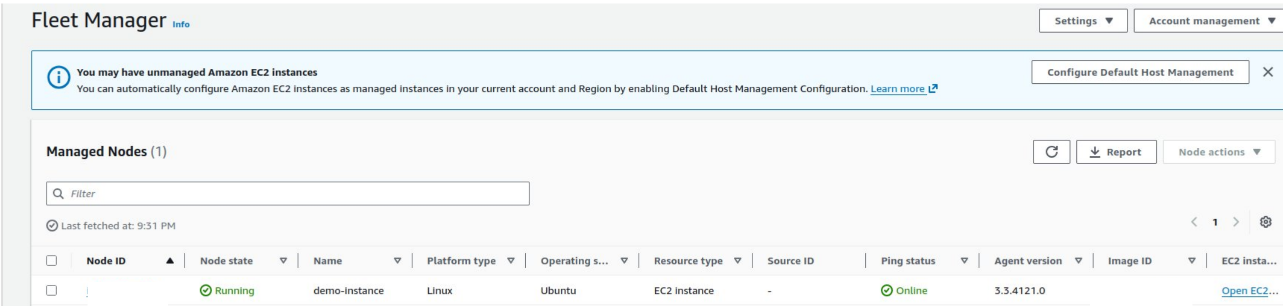


Step 4: Verify Systems Manager Registration

Verified that the EC2 instance was successfully registered in AWS Systems Manager Fleet Manager.

Screenshot 4

Systems Manager / Managed Node



Step 5: Execute Disk Utilization Command

Opened Systems Manager Run Command.

Selected document:
AWS-RunShellScript

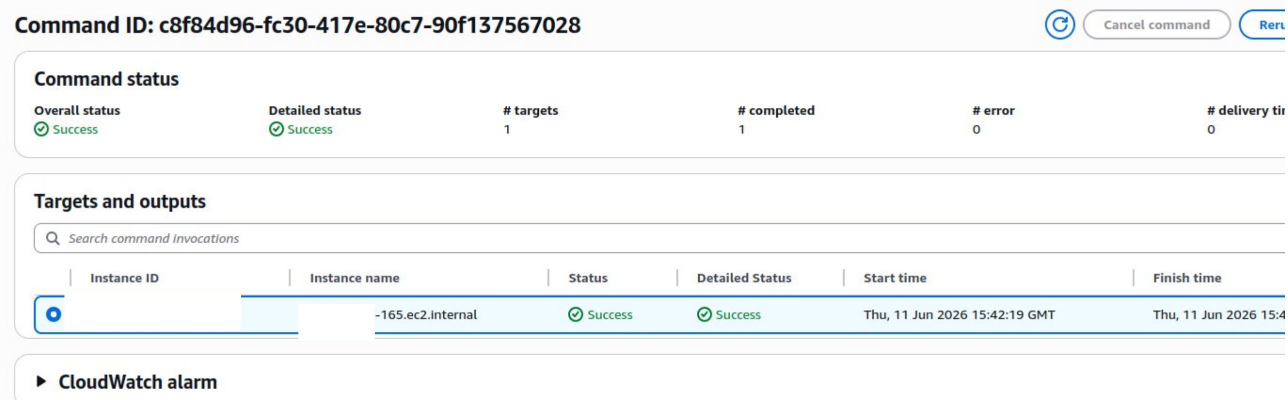
Executed command:

df -h

Screenshot 5

Run Command Success

[AWS Systems Manager Run Command executed successfully on the EC2 instance]

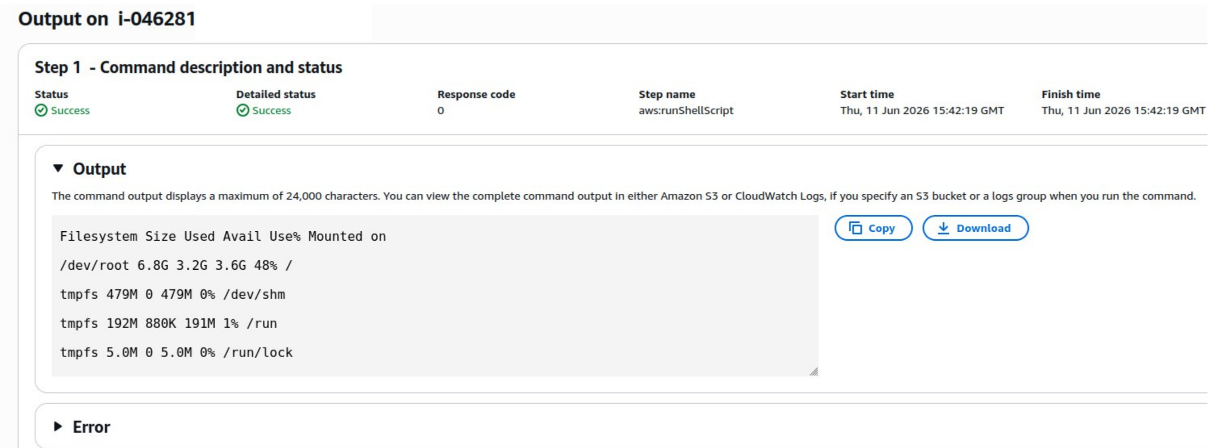


Step 6: Review Output

Collected disk utilization output returned by Systems Manager.

Screenshot 6

df -h Output



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6. Results and Analysis

Command Executed:

df -h

Output Summary:

Filesystem Size Used Avail Use% Mounted on
/dev/root 6.8G 3.2G 3.6G 48% /

/dev/xvda16 881M 152M 667M 19% /boot
/dev/xvda15 105M 6.1M 99M 6% /boot/efi

Analysis:

- Root Volume Size: 6.8 GB
- Used Space: 3.2 GB
- Available Space: 3.6 GB
- Disk Utilization: 48%

The root volume utilization is within acceptable limits and does not require immediate action.

7. Benefits of Using AWS Systems Manager

- No SSH or RDP access required.
- Improved security posture.
- Centralized management of EC2 instances.
- IAM-based access control.
- Ability to manage multiple instances simultaneously.
- Command execution history and auditing.
- Reduced operational overhead.

8. Limitations

- SSM Agent must be installed and running.
- IAM permissions must be configured correctly.
- Network connectivity to AWS Systems Manager endpoints is required.
- Primarily suitable for operational management rather than continuous monitoring.
- Additional monitoring solutions may be needed for long-term trend analysis.

9. Customer Communication Example

Scenario: High Disk Utilization

Subject: High Disk Utilization Detected on EC2 Instance

Dear Customer,

During our storage assessment, we observed that the EC2 instance is utilizing approximately 92% of its available storage capacity.

High disk utilization can lead to application performance issues, failed deployments, and service interruptions.

Recommended Actions:

- Remove unnecessary files and old backups.
- Configure log rotation for application logs.
- Increase the EBS volume size.
- Implement CloudWatch monitoring and alerts.
- Review storage usage patterns regularly.

Please let us know if you would like assistance implementing these recommendations.

10. Alternative Methods for Collecting Disk Utilization

Method 1: SSH Access

Administrators can connect directly to the EC2 instance using SSH and execute the df -h command to view disk utilization.

df -h

Method 2: Amazon CloudWatch Agent

The CloudWatch Agent can be installed on EC2 instances to collect disk utilization metrics and send them to CloudWatch.

Method 3: AWS Systems Manager Fleet Manager

Fleet Manager provides centralized visibility into managed instances and allows administrators to inspect instance details and perform management operations.